

Microcontroller-Based Multi-Sensor Spatial Sensing and Intelligent Navigation System

Abstract

Precision-guidance technologies, including 155 mm artillery ammunition equipped with Precision Guidance Kits (PGKs), demonstrate the importance of accurate positioning, sensing, and real-time computation in modern defence systems. This project explores the underlying concepts through a low-cost, modular multi-sensor spatial sensing and navigation prototype.

The proposed system integrates multiple advanced modules to achieve its navigation and sensing goals. GPS automatically determines the current geographical position, while a reference waypoint can be entered manually through a computer-based dashboard. The user can configure a search radius according to the available sensing hardware. Within this region, RF and infrared/thermal sensors perform angular and spatial scanning, collecting signal-strength and thermal-intensity measurements. The ESP32 filters the acquired data, identifies the strongest response regions, converts directional measurements into vectors, and applies a weighted sensor-fusion algorithm to obtain a resultant vector and navigation heading. If GPS becomes unavailable, the system can switch to a fallback mode using the last known position and local sensor information while reporting reduced confidence.

The software will use embedded C/C++ firmware and a Python-based dashboard, with algorithms and interfaces adaptable according to future requirements and hardware upgrades. Expected outputs include live GPS coordinates, waypoint information, sensor intensities, spatial heat maps, individual and fused vectors, estimated source location, confidence level, and navigation heading.

The system can support applications in autonomous robotics, search-and-rescue, disaster response, environmental monitoring, industrial inspection, infrastructure monitoring, and remote-area sensing.

Hardware Components:

- **ESP32 Microcontroller:** Handles data filtering, vector conversion, and sensor-fusion algorithms.
- **GPS Module:** Automatically determines the current geographical position.
- **RF Sensing Module:** Performs angular and spatial scanning to collect signal-strength measurements.
- **Infrared/Thermal Sensor:** Collects thermal-intensity measurements within the configured search radius.